Physics first, *code ever since.*

01 · MSC · 2017 – 2018

Imperial College London

Master of Science, Physics

DISSERTATION

Activation energies of ionic migration in perovskites, via Electrochemical Impedance Spectroscopy

Experimental physics + computation

Impedance measurements on perovskite solar cells, analysed and modelled in Python.

Wrote an EIS fitting algorithm

Fit equivalent-circuit models to impedance data by minimising mean-square error.

Results matched theory

Activation energies agreed with prior studies. Linear regression, Matplotlib visuals.

02 · BSC · 2011 – 2015

Universitas Gadjah Mada

Bachelor of Science, Condensed Matter & Materials Physics

UNDERGRADUATE THESIS

Electronic Transport in Semiconducting Graphene

Theory + computation

Python (NumPy, Matplotlib), Fortran and MATLAB.

Modelled a graphene pn junction

Two differently doped graphene layers, assessed for semiconductor devices.

Lengthy analytical derivations

Worked the maths through by hand, then solved and visualised it numerically.

Physics taught me to model messy data, then write the code that fits it.

London · Yogyakarta